

A First Project Proposal Report on

Kotha Chaiyo: A Web-Based Room Finder App

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ABSTRACT

“Kotha Chaiyo” is a user-friendly web application designed to simplify the process of finding suitable accommodations for rent. The primary objective of the platform is to facilitate the room/flat search process by offering advanced search filters and options. Users can easily search for rooms based on specific criteria such as location, price range, amenities, and room type. Landlord can directly list their property on the platform without any hassle. In the major cities like Kathmandu, the process of finding accommodation involves middle-men and agents who charge an extra amount which makes the process frustrating and time consuming. The platform intends to eliminate the middle-men and agents making the process efficient. Additionally, interactive map will be integrated to streamline the process of finding rooms so that it becomes easier for the tenants to find out the exact location of the property. The technologies like HTML, CSS, JavaScript will be used to build the frontend of the application. On the server side, PHP and MySQL will be used. With the advanced search feature and interactive maps, the process of finding accommodation gets simplified. This project will therefore simplify the process of finding accommodations solving the problems that exist in traditional approach.

Keywords: Landlord, Tenant, Agent, Admin, Interactive Map

TABLE OF CONTENTS

1. INTRODUCTION	1
1.1 PROBLEM STATEMENT	1
1.2 PROJECT OBJECTIVES	1
1.3 SCOPE AND LIMITATIONS.....	2
2. LITERATURE REVIEW	3
3. METHODOLOGY	4
3.1 SOFTWARE DEVELOPMENT LIFECYCLE.....	4
3.1.1 REQUIREMENT ANALYSIS	4
3.1.2 DESIGN.....	5
3.1.3 CODING	5
3.1.4 TESTING AND EVALUATION	5
3.2 TOOLS TO BE USED.....	5
3.3 TECHNOLOGIES TO BE USED	6
4. SYSTEM DESIGN	7
4.1 ENTITY-RELATIONSHIP (ER) DIAGRAM:	7
4.2 USE CASE DAIGRAM.....	8
5. PROPOSED DELIVERABLES	9
6. PROJECT TASK AND TIME SCHEDULE.....	10
6.1 GANTT CHART	10
7. BIBLIOGRAPHY	11

LIST OF FIGURES

Figure 1:Incremental Model of Software Engineering	4
Figure 2:Entity-Relationship (ER) Diagram	7
Figure 3: Use Case Diagram	8

LIST OF TABLES

Table 1: Time Schedule	10
Table 2:First Increment.....	10

1. INTRODUCTION

This project carries the motive of providing an efficient and convenient way to find and safeguard accommodations, no matter if it's for travellers, students or individuals searching for a new place to live. In this section, problems, motivations, objectives, project scope and limitations will be described in detail.

1.1 PROBLEM STATEMENT

In the present world, finding suitable accommodation can be a challenging task, whether it's for students, travellers or individuals looking for a new place to shift in. The conventional and traditional approach of finding rooms, such as ads or direct communication with landlords or agents, can be time-consuming and inefficient.

Also, it becomes difficult for people around the big cities like Kathmandu, Pokhara, Biratnagar etc. to find the exact room and accommodation they are looking for. The existing approach of finding rooms engages the involvement of middle-man and agents who charge extra amount from the clients for flats and rooms. Additionally, scams and frauds are recorded with the individual searching for the rooms from the agents and middleman.

This can lead to frustration, wastage of effort and money, and difficulties in finding suitable accommodations within desired requirements.

1.2 PROJECT OBJECTIVES

To address the problems stated above in problem statement section 1.1, the existing and traditional approach of finding accommodations should be completely reformed. So, with the purpose of providing an efficient solution for the stated problems, Kotha Chaiyo web-app is introduced with the following objectives:

- i. Simplify the room search process: The primary objective of this project is to provide an efficient user-friendly platform for users to find rooms based on their requirements, locations, price and amenity.
- ii. Provide an interface for communication between landlord and tenant: Functioning as a modern web-app, it allows users to communicate with landlords to interact with the terms like inquiries, negotiations, site viewing etc.

- iii. Increase efficiency and save time: Unlike conventional methods, tenants do not have to physically interact with landlords about the rooms and flats. Also, the app provides a complete platform for users so that they do not have to visit multiple websites and contact multiple landlords.
- iv. Eliminate the middle-men and agents: The traditional approach involves the middle-man and agents, which makes the process more complicated and frustrating. Also, most of the frauds and scams are recorded by the middle-men and agents. So, this app focuses on eliminating them and making the process easier and efficient.

1.3 SCOPE AND LIMITATIONS

In the project ‘Kotha Chaiyo’, a system will be developed in such a way that user and landlord can share a centralized platform where they can communicate with each other and can make a deal online.

The scopes of this project are:

- i. The target audience can be anyone who is looking for a room/flat and the flat owners who want to rent their property.
- ii. It especially targets the students, travellers or an individual who is looking for a flat/room to live in.
- iii. By integrating the maps, the proposed web-app can be used in any part of Nepal.

The limitations of this project are:

- i. It is limited to specific locations (Kathmandu) as of now.
- ii. The users must be digitally literate as it involves the online approach of finding and renting rooms/flats.

2. LITERATURE REVIEW

In this section, we have reviewed some of the existing platforms providing the similar service that are currently in use.

The first site that we found was propertynepal.com. It is a real-estate buy, sell and rent platform that allows people to buy, sell or rent properties. One has to create an account first to get started with the platform. He/she can register as a user or an agent where the user can get the information about the listed properties and agents can list the properties on the platform. The major drawback we found about this platform is that it involves agents who would charge an extra amount from the tenant. Additionally, the site does not guarantee the genuineness/accuracy of the information posted. This could lead to fraudulent activities and scams. In our proposed idea, we will be implementing a solution to monitor users if they are involved in any fraudulent activities or if there are any agents who are posting a number of listings at the same time.

Another platform with the name roomsfindernepal.com also provides similar service. However, the platform contains old listings, some listings that were posted years ago also exist in the platform and have not been removed yet. This could lead to frustration to a user because the platform does not properly monitor the property listings so a user wouldn't know if the property has already been rented. In our project, we would be implementing a system to notify and ask the landlord if his property listed on the platform has been rented or not, after a certain time interval so that landlord would be able to remove the listing. This way, a user would be able to conveniently find the property s/he is looking for.

3. METHODOLOGY

We will be adopting the following methodologies for a smooth development process to meet the requirements of the project and not deviate from it.

3.1 SOFTWARE DEVELOPMENT LIFECYCLE

We will be following the incremental model of software development as a framework for this project. The incremental model of software development is an approach where the development process is broken down into smaller increments, each involving different phases such as Requirement Analysis, Design, Coding, Testing and Evaluation. In this model, each increment delivers a partial but usable version of the project.

In the first increment, we will be developing the core product having major functionalities such as user account creation, login, property listing functionality for room/flat owners, property viewing and interaction functionality for the room finders, integration of map for precise location.

In the subsequent increments, we will be adding more functionalities to satisfy the requirement specification. Each increment has following phases:

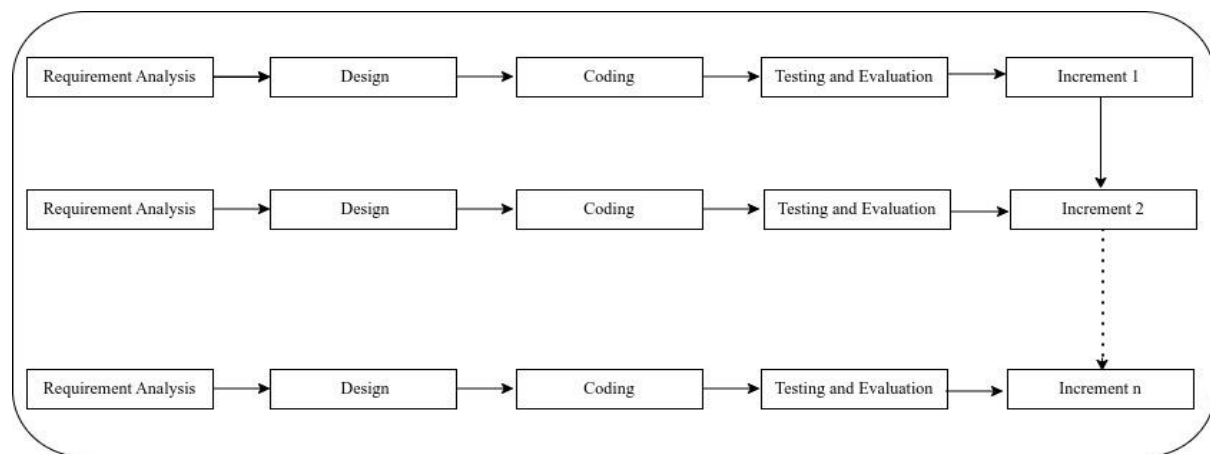


Figure 1: Incremental Model of Software Engineering

3.1.1 REQUIREMENT ANALYSIS

This is the first step of an incremental model where requirements will be collected and analysed. The outcome of this phase will be a System Requirement Specification (SRS) document.

3.1.2 DESIGN

In this phase, we will be designing the system based on the System Requirement Specification (SRS). We will be developing ER diagrams and Use Case diagrams in this phase.

3.1.3 CODING

The functionalities as specified in the System Requirement Specification (SRS) and as modelled during the design phase, will be developed through code in this phase. The outcome of this phase will be a working product with the functionalities as specified in the document.

3.1.4 TESTING AND EVALUATION

In this phase, different modules of the system will be tested individually. The entire system will also be tested to assure that the functionalities provided by the system work smoothly as specified.

3.2 TOOLS TO BE USED

We will be using the following tools throughout this project to streamline the development process.

TRELLO:

Trello is a web-based project management tool that we will be using to collaborate with each other and to organize and manage individual tasks.

GIT AND GITHUB:

Git is a distributed version control system that will be used in this project to manage different versions of the project and to collaborate with each other. GitHub is a platform that uses git for version control. We will be using GitHub to host the project repository, to track changes and manage different versions of the project.

VS CODE:

VS Code is a code editor that will be used as a primary code editor for development tasks.

XAMPP:

XAMPP is an open-source software package that provides a local server environment for running dynamic web applications on a personal computer. We will be using it for building and testing our web application.

3.3 TECHNOLOGIES TO BE USED

We will be using the following technologies during the project development

HTML:

HTML stands for Hypertext Mark-up Language. It is a standard mark-up language that defines the structure of a webpage. It consists of tags that wrap the content based on which the content is displayed on the webpage. We will be using HTML as a mark-up language for the project.

CSS:

CSS stands for Cascading Stylesheets. It is a stylesheet language that is used to style the components of a webpage. CSS will be used to style the components written in HTML.

JAVASCRIPT:

JavaScript is a client-side scripting language used to write client-side logic and validations. JavaScript will be used for front-end development.

PHP:

PHP stands for “PHP: Hypertext Preprocessor”. It is a server-side scripting language used to write server-side logic and generate dynamic content. We will be using PHP as a scripting language for the backend of the project.

MYSQL:

We will be using SQL (Structured Query Language) as a query language to interact with the database. MySQL will be used as a Relational Database Management System (RDBMS) for the project.

4. SYSTEM DESIGN

This section provides a comprehensive overview of the proposed system's architecture focusing on two key diagrams: Entity-Relationship (ER) diagram and the Use Case diagram. These diagrams illustrate the core components and the interactions between them on the system.

4.1 ENTITY-RELATIONSHIP (ER) DIAGRAM:

An Entity-Relationship (ER) diagram is a visual representation of the entities(objects) within a system and the relationship between them. It is commonly used in database design to model the structure and the organization of data. The major entities involved in our system are Tenant, Landlord, Property and the Admin. The relationship between these entities is shown in the given ER diagram.

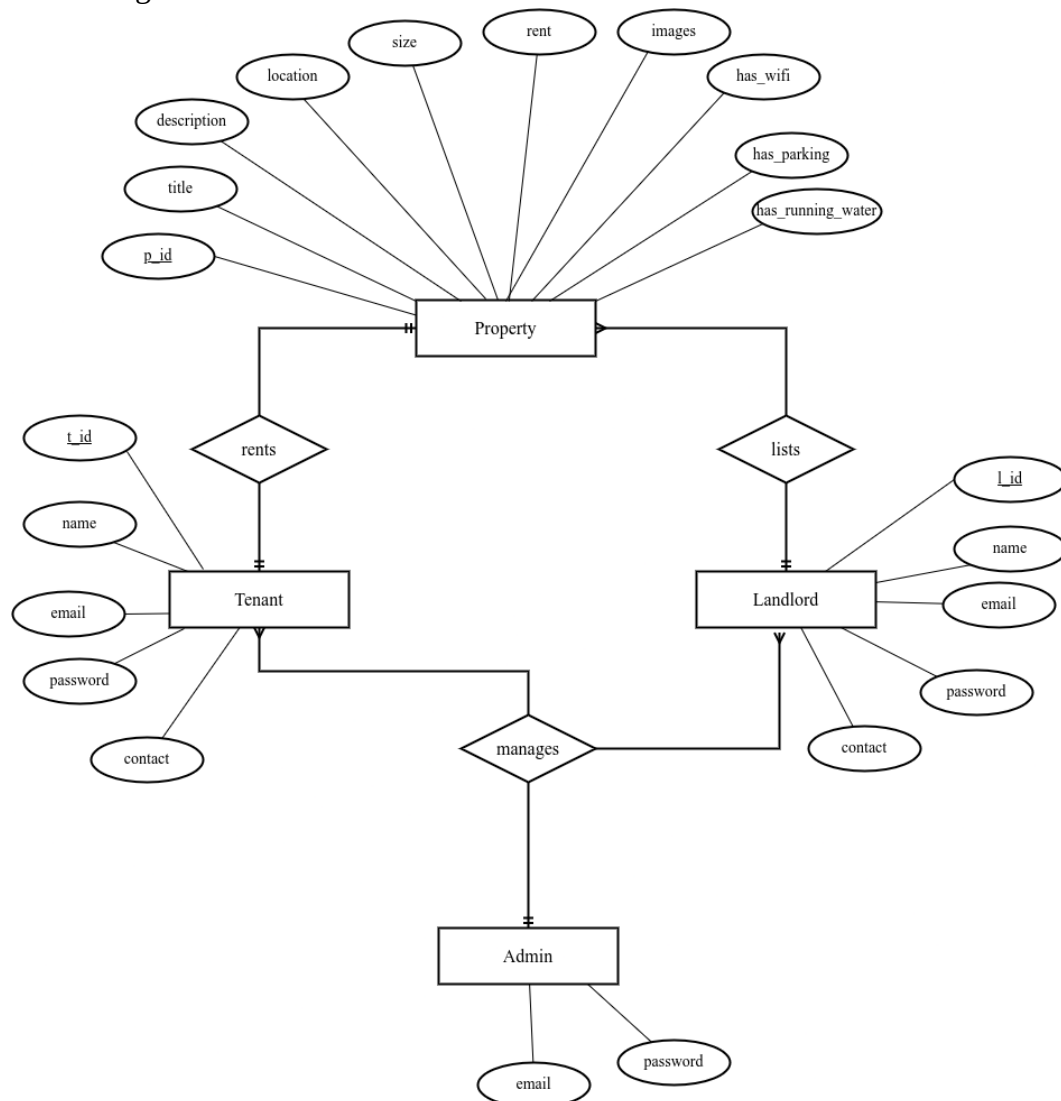


Figure 2:Entity-Relationship (ER) Diagram

4.2 USE CASE DAIGRAM

A use case diagram is a visual representation of the interactions between actors (In our case: Landlord, Tenant and Admin) and a system under consideration. It shows the different use cases or functionalities provided by the system and the relationships between the actors and the use cases.

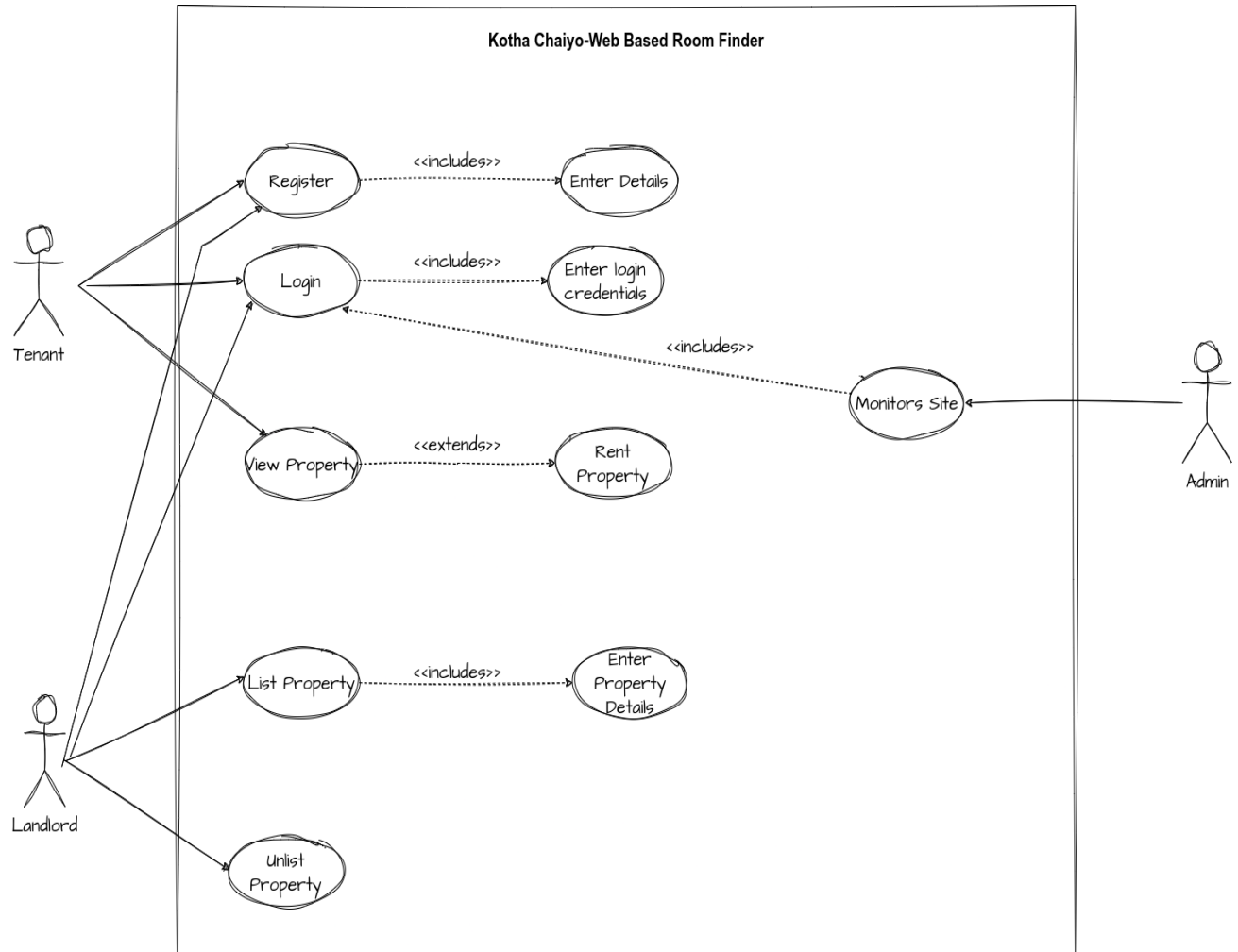


Figure 3: Use Case Diagram

5. PROPOSED DELIVERABLES

The project will deliver a responsive platform independent Web Application. The user will be able to use and utilize the application via a web browser in platform of their preference. There are essentially three users of our application. The Landlord, one who puts room for rent, The Tenant, one who searches for room and The Admin who monitors the application for various fraud and banns user.

In the application, the tenant is able to search for rooms that are currently in rent. The tenant can also filter the rooms based on various preferences.

The Land Lord will be able to put-up various rooms/flats in rent. They can put-up photos, location, and description of their property.

The Admin will have statistics of the user and by analysing user pattern the admin will be able to ban and moderate application.

6. PROJECT TASK AND TIME SCHEDULE

The project time schedule is designed, considering the requirements and constraints involved. The project schedule is designed such that the Requirement Analysis phase is given more importance so that we don't deviate from the requirement of the project.

TASK	APPROX DURATION IN DAYS
Requirements analysis	10
System Design	5
Coding	15
Testing and Debugging Individual Modules	4
Overall System Test and Debugging	3
Documentation	7

Table 1: Time Schedule

6.1 GANTT CHART

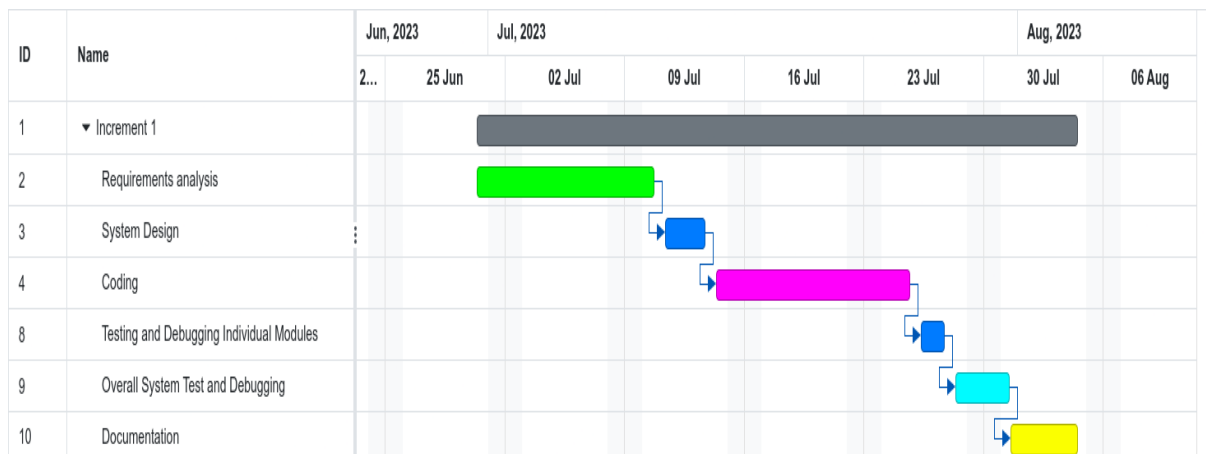


Table 2: First Increment

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- <https://www.php.net/docs.php>